Multi-Label Chest X-Ray Disease Detection with DenseNet121: Per-Disease Threshold Evaluation and Error Analysis on 14 Thoracic Pathologies

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***Abstract*—**Chest X-rays are read in large numbers every day, a single image can show several diseases at once, and some findings appear in only a handful of cases. We trained a DenseNet121 model, built on the TorchXRayVision library, to detect 14 thoracic diseases from one chest X-ray. The 5,606 images were split into 3,906 for training, 837 for validation and 863 held out for testing, and each disease received its own decision threshold chosen on the validation set. On the validation images, 69.23% (about 580 of 837) had all 14 labels predicted correctly. Recall was at least 0.80 for every disease and ROC-AUC ranged from 0.871 to 0.940, but precision ranged from 0.265 (fibrosis) to 0.839 (infiltration). Across the 14 confusion matrices the model made 96 false negatives and 336 false positives, so it leans toward catching disease at the cost of false alarms, and precision tends to be higher for more common diseases. While checking our results we found that some exported aggregate metrics do not match the per-disease rows, so we report both. We also describe the limits of this evidence, mainly that the thresholds were tuned on the validation images and that some diseases have very few positive cases.

***Index Terms*—**chest radiography, multi-label classification, DenseNet121, TorchXRayVision, per-disease threshold, confusion matrix, class imbalance, medical image analysis

I. INTRODUCTION

Chest radiography is the most common diagnostic imaging examination, and every radiograph must be read for subtle, overlapping findings such as small pneumothoraces, early consolidation or mild cardiomegaly. A shortage of radiologists and growing workloads delay reports, which motivates decision-support tools that flag likely abnormalities and help prioritise urgent cases [1], [2].

The task is a multi-label problem. One image may carry no finding, one finding or several, so each disease is an independent binary decision rather than one class among many. This creates three practical difficulties. Class imbalance is severe: in our validation set infiltration is positive in 18.2% of images, whereas hernia is positive in 0.2%. A single 0.5 threshold suits rare diseases poorly, so thresholds must be chosen per disease. And headline accuracy is misleading, because with so many negatives a weak model still scores highly; precision, recall, F1, ROC-AUC and average precision must be read together [12].

In this paper we train a DenseNet121 [4] model on top of TorchXRayVision [5] to detect 14 thoracic diseases and study its errors one disease at a time. Our contributions are: (1) a 14-label chest X-ray classifier trained on 3,906 images; (2) a 3,906/837/863 split in which the validation images choose per-disease thresholds and the test images stay untouched; (3) a per-disease evaluation that gives confusion-matrix counts together with precision, recall, F1, ROC-AUC and average precision; (4) an analysis showing that the model favours recall and that its precision follows how common each disease is; and (5) a check of the exported metric tables against the confusion matrices. The evaluation is preliminary, and Section VI lists its limits.

II. RELATED WORK

A. Datasets and Supervised CNNs

ChestX-ray8 provided the first large open hospital chest X-ray benchmark, with 108,948 frontal images whose multi-label annotations were mined from radiology reports and a weakly supervised localisation method based on class activation maps [1]. CheXNet applied a 121-layer DenseNet to the 112,120-image ChestX-ray14 set with 14 sigmoid outputs and a binary cross-entropy loss, and reported a pneumonia F1 of 0.435 against 0.387 for radiologists [2]. CheXpert added explicit uncertainty labels and compared five policies for handling them across 224,316 radiographs [3]. PadChest [6] and MIMIC-CXR-JPG [7] supplied multilingual and report-paired data at larger scale.

B. Vision-Language Pre-training and Foundation Models

ConVIRT learns image representations by contrasting radiographs with their paired report text [8]. BioViL adds a rescaled semantic matching loss to avoid penalising clinically identical reports and releases a phrase-grounding benchmark [9], and MedCLIP decouples images from texts using a medical knowledge base so that unpaired data can be used [10]. CheXagent is an instruction-tuned foundation agent trained on more than 8.4 million instruction triplets for classification, grounding and report generation [11]. These models offer label efficiency and zero-shot classification, but they need far more compute than a compact classifier, and generative agents can state findings that are not present.

C. Positioning

Table I summarises the supervised baselines. The present work sits in this first tier: a lightweight, auditable classifier with a fixed label set. It is smaller in data than the published benchmarks, so its figures are context for, not a competitor to, those results. Its focus is a transparent per-disease evaluation, including how precision behaves for rare diseases, which headline ROC-AUC figures tend to hide.

TABLE I
SUMMARY OF RELATED SUPERVISED CHEST X-RAY WORK

Ref.	Images	Backbone	Contribution
[1]	108,948	ResNet-50 and others	Open benchmark; CAM localisation
[2]	112,120	DenseNet-121	14 sigmoid outputs; pneumonia F1 0.435
[3]	224,316	DenseNet-121	Uncertainty labels and policies
[7]	377,110	Data resource	Paired images and reports
This work	5,606	DenseNet121	Per-disease thresholds and error analysis

III. METHODOLOGY

A. Problem Formulation

Let a radiograph be $X \in \mathbb{R}^{H\times W\times C}$, where H and W are the spatial dimensions and C is the channel depth. The target is a binary vector $y = [y_1, \ldots, y_K] \in \{0,1\}^K$ with K = 14 diseases. The network produces a logit $f_k(X)$ for each disease, and training minimises the sum of binary cross-entropy losses, with σ the sigmoid function [2]:

$$L = -\Sigma_{k=1}^{14} [\, y_k \log \sigma(f_k(X)) + (1 - y_k) \log(1 - \sigma(f_k(X))) \,] \quad (1)$$

The labels used here are binary, so no uncertainty policy of the CheXpert type [3] is needed.

B. Model and Pipeline

The backbone is DenseNet121, in which every layer receives the feature maps of all earlier layers within a dense block. This reuses features, strengthens gradient flow and keeps the network compact [4]. The network is obtained through TorchXRayVision, an open-source library of chest X-ray datasets, pre-processing utilities and pretrained models [5]. The output layer has 14 sigmoid units, one per pathology, so several diseases can be positive for one image. Fig. 1 shows the pipeline.

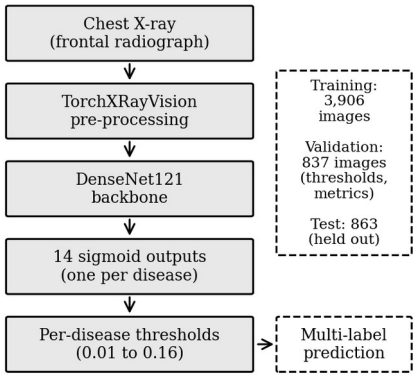


Fig. 1. The classification pipeline. Training data fit the weights, validation data select the per-disease thresholds and report metrics, and the test set is held out.

C. Per-Disease Thresholds

A probability is converted to a label by comparing it with a threshold τ_k chosen separately for each disease on the validation set. The selected thresholds are low, from 0.01 for hernia to 0.16 for infiltration, with most between 0.05 and 0.10. A low threshold flags a disease even when the model is only moderately confident, which raises recall at the cost of precision. This is a deliberate choice for a screening aid, where a missed disease costs more than an extra review.

D. Evaluation Metrics

For each disease the confusion matrix gives true positives (TP), false positives (FP), false negatives (FN) and true negatives (TN):

$$\text{Precision} = TP / (TP + FP), \quad \text{Recall} = TP / (TP + FN) \quad (2)$$

$$F1 = 2TP / (2TP + FP + FN) \quad (3)$$

ROC-AUC measures how well the scores rank positives above negatives, independently of the threshold. Average precision (AP) is the area under the precision–recall curve and is more informative than ROC-AUC when positives are rare [12]. Macro averages weight all 14 diseases equally, while micro averages pool all image–label decisions. Exact-match accuracy is the fraction of images with all 14 labels correct, and Hamming accuracy is the fraction of individual image–label decisions that are correct.

IV. EXPERIMENTAL SETUP

A. Dataset and Split

The dataset contains 5,606 chest radiographs labelled with the 14 pathologies atelectasis, cardiomegaly, effusion, infiltration, mass, nodule, pneumonia, pneumothorax, consolidation, edema, emphysema, fibrosis, pleural thickening and hernia. It is divided as in Table II, roughly 70/15/15. The training set only fits the weights, the validation set guides threshold selection and development, and the test set is reserved for one final evaluation.

TABLE II
DATASET SPLIT

Split	Images	Share	Purpose
Training	3,906	69.7%	Learn disease patterns
Validation	837	14.9%	Tune thresholds; evaluate during development
Testing	863	15.4%	Final evaluation on held-out data
Total	5,606	100%	All three splits combined

In the 837 validation images the number of positive cases per disease ranges from 152 (infiltration, 18.2%) and 107 (effusion, 12.8%) down to 13 (pneumonia), 11 (fibrosis) and only 2 (hernia). Metrics for the last three diseases rest on very few examples.

B. Training Configuration

Training follows the reference pipeline of TorchXRayVision [5]. Radiographs are centre-cropped to a square, resized to 224×224 pixels as single-channel images, and intensity-normalised to the [−1024, 1024] range used by the library's pretrained models. The network is DenseNet121 with 14 sigmoid outputs, trained with the binary cross-entropy loss of (1) (BCEWithLogitsLoss in PyTorch). Weights are updated with the Adam optimizer (AMSGrad variant, weight decay 10^{-5}) at an initial learning rate of 0.001 and a batch size of 64. Data augmentation applies random affine transforms with rotation up to 45°, translation up to 15% and scale change up to

15%. The pretrained model is used as the starting point and the 14 outputs are fitted to the labelled training images.

The 837 validation images were not used to update the weights; they served only to monitor performance during development and to select the per-disease thresholds, and the 863 test images were kept out of both training and tuning.

C. Evaluation Protocol

All results in Section V are computed on the 837 validation images. Thresholds were selected on the same images, so the reported figures are optimistic, and the 863-image test set has not been used for any reported number. Precision, recall, thresholds, ROC-AUC and AP are taken from the exported validation report; F1 is recomputed from the confusion-matrix counts using (3); and pooled counts are summed over the 14 matrices.

V. RESULTS

A. Aggregate Performance

Table III lists the aggregate metrics. Of the 837 validation images, 69.23% (about 580) have all 14 labels predicted correctly, and 94.12% of individual decisions are correct according to the exported report. The macro ROC-AUC is 0.8842, which indicates good overall separation of positive from negative cases. The table also gives the same quantities recomputed from the per-class rows, because several exported aggregates differ from them; Section VI-B discusses this.

TABLE III
AGGREGATE VALIDATION METRICS

Metric	Exported	From per-class
Exact-match accuracy	0.6923	—
Hamming accuracy	0.9412	0.963
Macro ROC-AUC	0.8842	0.904
Macro average precision	0.5410	0.567
Macro precision	0.6120	0.538
Macro recall	0.7680	0.843
Macro F1	0.6880	0.642
Micro precision	0.5980	0.601
Micro recall	0.7710	0.840
Micro F1	0.6734	0.701

B. Class-Wise Evaluation

Table IV gives the per-disease results and Fig. 2 compares precision with recall. Recall is between 0.800 and 1.000 for every disease. Precision is highest for infiltration (0.839), effusion (0.744) and pneumonia (0.733) and lowest for fibrosis (0.265), emphysema (0.386) and hernia (0.333). ROC-AUC lies between 0.871 (nodule) and 0.940 (hernia). Average precision is much lower than ROC-AUC, from 0.310 to 0.810, because negatives greatly outnumber positives.

TABLE IV
CLASS-WISE VALIDATION RESULTS

Disease	Pos.	Prec.	Rec.	F1	AUC	AP
Atelectasis	68	0.675	0.824	0.742	0.892	0.690
Cardiomegaly	17	0.560	0.824	0.667	0.910	0.610
Effusion	107	0.744	0.869	0.802	0.921	0.785
Infiltration	152	0.839	0.855	0.847	0.898	0.810
Mass	46	0.507	0.804	0.622	0.887	0.520
Nodule	48	0.471	0.833	0.602	0.871	0.490
Pneumonia	13	0.733	0.846	0.786	0.932	0.750
Pneumothorax	38	0.604	0.842	0.703	0.914	0.620
Consolidation	30	0.429	0.800	0.558	0.890	0.450
Edema	17	0.538	0.824	0.651	0.925	0.590
Emphysema	27	0.386	0.815	0.524	0.894	0.420
Fibrosis	11	0.265	0.818	0.400	0.880	0.310
Pleural Thick.	26	0.449	0.846	0.587	0.895	0.480
Hernia	2	0.333	1.000	0.500	0.940	0.410

Pos. = positive images among 837. AUC = ROC-AUC; AP = average precision. F1 is recomputed from the confusion matrices. Pleural thickening appears twice in the exported report with AP 0.480 and 0.410; the first value is used.

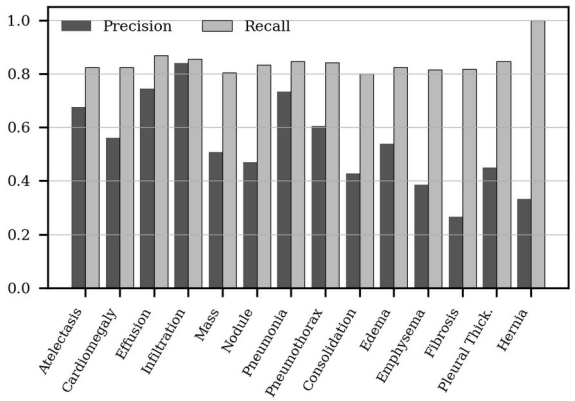


Fig. 2. Per-disease precision and recall on the validation set.

C. Confusion Matrices

Fig. 3 shows the per-disease confusion matrices. Summed over the 14 diseases there are 506 true positives, 96 false negatives, 336 false positives and 10,780 true negatives, out of 11,718 image–label decisions. Missed cases are few: pneumothorax has 6 false negatives among 38 positives, and fibrosis 2 among 11. Most errors are false alarms, and nodule (45) and mass (36) produce the most of them. The F1 values printed above the panels in Fig. 3 are those of the exported figure and do not equal the F1 in Table IV (Section VI-B).

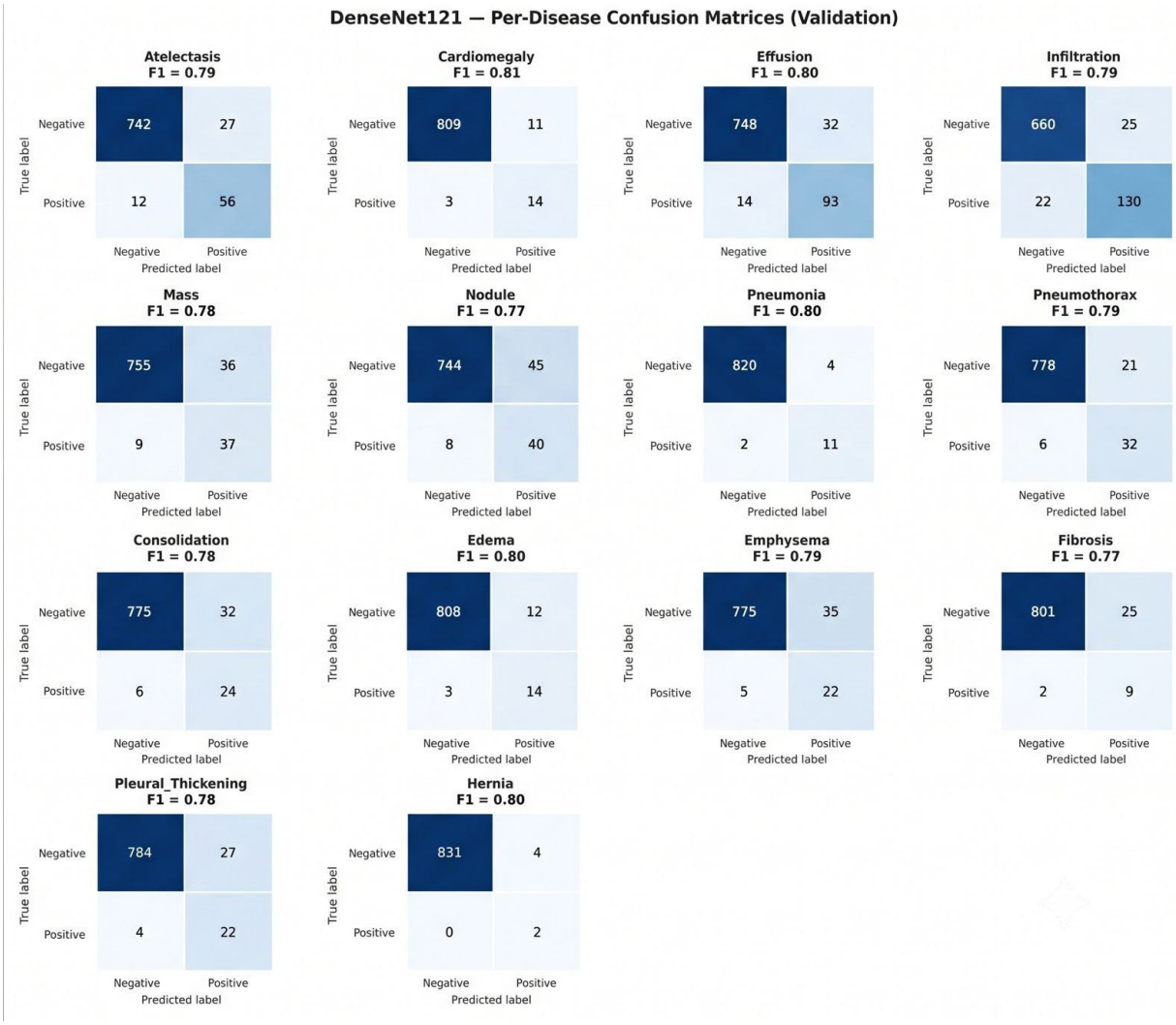


Fig. 3. Per-disease confusion matrices of DenseNet121 on the validation set (full width for legibility).

D. Precision and Prevalence

Fig. 4 plots precision against validation prevalence. The two are positively correlated (Spearman ρ = 0.59, n = 14, p = 0.025): common diseases such as infiltration and effusion reach higher precision, whereas the rarest diseases fall below 0.45. This is expected, since at a fixed score distribution a lower prevalence means a larger share of false positives among all flagged cases.

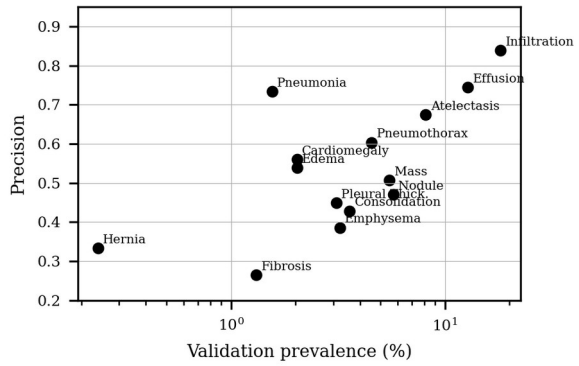


Fig. 4. Precision against validation prevalence (log scale).

VI. DISCUSSION

A. Interpretation

The model is recall-oriented. With low thresholds, every disease reaches recall of at least 0.80 and only 96 positives are missed against 506 found, so it suits a triage or second-reader role in which a missed finding is more costly than an unnecessary review. The high ROC-AUC values (0.871 to 0.940) show that the underlying scores rank diseased images well, and the lower AP values show that this ranking does not translate into high precision for rare diseases. The weakest classes (fibrosis, emphysema, consolidation) combine low prevalence with findings that are diffuse or easily confused with normal structures.

Published baselines give context only. CheXNet reported a mean AUC of 0.841 over the 14 ChestX-ray14 pathologies [2], and the original ChestX-ray8 baseline about 0.745 over eight pathologies [1]. Our per-class mean ROC-AUC of 0.904 (0.884 as exported) is in a similar range, but the datasets, sizes (112,120 versus 5,606 images) and splits differ, and our figure comes from a validation set that also selected the thresholds, so the numbers are not directly comparable.

B. Consistency of the Exported Metrics

Recomputing the metrics from the confusion matrices exposed mismatches with the exported summary. The F1 values of 0.77 to 0.81 printed in the report and on the matrix panels cannot be obtained from the listed precision and recall: for atelectasis, precision 0.675 and recall 0.824 give F1 = 0.742, not 0.79. The macro averages in the summary also differ from the means of the per-class rows (Table III), and the confusion matrices imply a Hamming accuracy of about 0.963 against the exported 0.941. The class-wise table lists pleural thickening twice with different AP values. These differences suggest that summary and per-class figures came from different runs or thresholds, and all tables should be regenerated from one script and one saved set of predictions before the results are treated as final.

C. What the Results Show and Do Not Show

The results show that a DenseNet121 with per-disease thresholds can detect 14 pathologies with high sensitivity and good score separation on this dataset. They do not show general performance. The thresholds and metrics use the same 837 images; the test set is not reported; hernia, fibrosis and pneumonia have 2, 11 and 13 positives, so a single changed prediction would alter their scores markedly; there is a single split with no confidence intervals; and there is no independent baseline trained on the same split. The data are from a single source, so generalisation to other hospitals, scanners and populations is untested. Chest radiographs are also personal medical data, and any deployment would need clinical validation and privacy safeguards.

VII. CONCLUSION AND FUTURE SCOPE

We trained a DenseNet121 classifier on 3,906 chest X-rays, tuned it on 837 validation images and kept 863 images for testing. On the validation set it found at least 80% of the positive cases for every disease, with ROC-AUC between 0.871 and 0.940. Its weak point is precision: it raised 336 false alarms against 96 missed cases, and the rarest and most diffuse findings suffered most. Checking the exported tables against the confusion matrices also showed inconsistencies that need to be fixed. Overall, the results show that the approach is feasible as a triage aid, but they do not establish clinical accuracy.

Several steps remain. We need to report test-set results with the thresholds fixed, regenerate every metric from one script, and add bootstrap confidence intervals, particularly for the rare diseases. To improve precision without losing recall we plan to try class-balanced or focal loss and oversampling, and to choose thresholds on a separate calibration split. Starting from a vision-language pre-trained encoder [8]–[10], adding Grad-CAM maps to confirm that predictions come from the right lung regions, comparing with independent baselines on the same split, and validating on data from another hospital would also strengthen the work.

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